

GURAYAT, Saudi Arabia

WMO: 403600

Lat: 31.412N

Lon: 37.279E

Elev: 510

StdP: 95.35

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	1.2	-12.9	1.3	15.7	-10.1	1.7	15.9	12.8	11.3	10.6	11.5	1.4	270	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	41.1	18.3	39.9	18.3	38.2	17.9	22.0	33.4	20.8	32.1	20.0	31.3	4.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.5	27.4	17.2	13.1	25.2	16.2	12.2	24.4	67.3	32.8	62.6	32.1	59.3	31.0	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
11.5	10.2	9.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-3.5	43.8	2.4	1.3	-5.2	44.8	-6.6	45.6	-8.0	46.3	-9.7	47.3
			-4.9	23.0	2.4	2.3	-6.6	24.6	-8.0	25.9	-9.3	27.2	-11.0	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	8.9	11.2	15.3	20.0	24.8	27.8	29.9	29.8	27.1	22.5	15.5	10.3	(6)
(7)		DBStd	8.05	2.50	2.83	3.57	3.98	3.49	2.55	2.13	2.19	2.52	3.23	3.37	2.65	(7)
(8)		HDD10.0	99	49	17	2	0	0	0	0	0	0	3	28	(8)	
(9)		HDD18.3	979	292	201	109	27	1	0	0	0	5	96	248	(9)	
(10)		CDD10.0	3862	16	50	165	301	460	535	616	614	513	388	167	38	(10)
(11)		CDD18.3	1700	0	0	14	77	202	285	357	356	263	135	10	0	(11)
(12)		CDH23.3	22798	0	14	241	1004	2695	3954	5084	4927	3310	1443	125	2	(12)
(13)		CDH26.7	13428	0	1	77	431	1479	2437	3287	3153	1918	626	20	0	(13)
(14)	Wind	WSAvg	4.2	3.0	3.5	4.2	4.4	4.7	5.5	5.9	5.5	4.6	3.4	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	50	12	6	8	4	2	0	0	0	4	3	11	(15)	
(16)		PrecMax	135	73	26	35	36	27	0	0	1	2	51	14	55	(16)
(17)		PrecMin	14	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	29	15	6	11	9	5	0	0	0	0	11	4	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	25.2	32.6	36.7	40.1	42.0	43.2	43.0	41.1	36.1	29.0	22.8	(19)
(20)			MCWB	10.7	11.6	14.4	16.2	17.0	17.9	19.2	20.1	18.4	17.4	14.3	12.0	(20)
(21)		2%	DB	18.1	22.1	28.9	33.5	37.8	40.0	41.1	40.9	38.6	34.1	26.8	19.9	(21)
(22)			MCWB	9.8	10.8	13.3	14.8	16.5	17.4	18.1	19.0	18.0	16.8	14.0	11.0	(22)
(23)		5%	DB	16.4	19.9	25.9	31.1	35.9	38.1	39.4	39.2	36.8	32.1	24.5	18.0	(23)
(24)			MCWB	9.1	10.2	12.2	14.1	15.9	17.4	18.3	18.7	17.6	16.3	13.3	10.3	(24)
(25)		10%	DB	15.0	18.0	23.2	29.0	34.0	36.5	38.0	37.9	35.0	30.2	22.6	16.3	(25)
(26)			MCWB	8.5	9.5	11.2	13.7	15.5	17.2	17.9	18.4	17.4	16.0	12.8	9.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	13.2	15.6	17.3	19.8	23.5	23.7	24.3	22.4	19.9	16.7	13.7	(27)
(28)			MCDWB	17.3	20.3	27.1	29.6	33.7	35.6	36.8	35.7	31.9	29.5	23.8	19.6	(28)
(29)		2%	WB	10.6	11.9	14.1	16.1	18.0	20.2	21.6	22.0	20.6	18.4	15.5	11.9	(29)
(30)			MCDWB	16.3	19.9	25.4	29.8	32.5	32.9	34.1	32.8	30.6	28.8	22.6	17.8	(30)
(31)		5%	WB	9.7	10.9	12.9	15.0	17.0	18.8	20.2	20.9	19.5	17.5	14.6	10.9	(31)
(32)			MCDWB	15.1	18.0	23.7	28.7	31.9	32.7	32.1	31.8	30.1	27.9	21.6	16.8	(32)
(33)		10%	WB	9.0	10.0	11.9	14.0	16.2	17.9	19.4	20.0	18.7	16.8	13.7	9.9	(33)
(34)			MCDWB	14.3	16.8	22.3	27.6	31.2	32.1	32.2	31.1	29.8	27.0	20.7	15.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.3	13.2	14.3	15.4	16.3	17.1	16.9	16.7	16.1	14.9	13.6	12.7	(35)
(36)			MCDBR	14.4	16.0	17.9	17.7	17.9	18.3	17.7	17.3	17.4	16.5	15.4	14.4	(36)
(37)			MCWBR	8.0	7.7	7.4	6.6	5.9	4.8	4.4	4.2	4.7	5.7	7.0	8.1	(37)
(38)		5% WB	MCDWB	12.9	14.0	15.9	16.5	16.3	17.2	16.8	16.9	15.8	14.3	12.6	12.7	(38)
(39)			MCWBR	7.7	7.4	7.1	6.7	5.9	5.5	4.9	4.7	4.7	5.2	6.0	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.376	0.424	0.485	0.483	0.408	0.410	0.415	0.408	0.436	0.383	0.354	(40)	
(41)		taud	2.237	2.146	1.994	1.843	1.875	2.114	2.128	2.120	2.124	2.028	2.146	2.221	(41)	
(42)		Ebn at Noon	862	871	853	815	817	878	874	866	858	795	811	831	(42)	
(43)		Edh at Noon	113	137	171	208	202	159	156	155	148	152	123	110	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.42	4.30	5.63	6.68	7.57	8.35	8.06	7.38	6.39	4.97	3.80	3.19	(44)	
(45)		RadStd	0.15	0.26	0.22	0.22	0.31	0.09	0.15	0.14	0.18	0.19	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	N/A
		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+143	+95

Nomenclature: See separate page